

# Joontaek Oh

Postdoctoral Researcher, School of CS, UW-Madison, USA

Email: [joontaek.oh@wisc.edu](mailto:joontaek.oh@wisc.edu)

LinkedIn: [linkedin.com/in/joontaek](https://www.linkedin.com/in/joontaek)

GitHub: [github.com/joontaekoh](https://github.com/joontaekoh)

Website: [joontaekoh.github.io](https://joontaekoh.github.io)

Google Scholar: [scholar.google.com/citations?user=WKX0HNsAAAAJ](https://scholar.google.com/citations?user=WKX0HNsAAAAJ)

## RESEARCH INTEREST

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Operating System, Filesystem, Storage system, Flash Storage, Database systems, Manycore scalability

## EDUCATION

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**Korea Advanced Institute of Science and Technologies (KAIST)** Daejeon, Korea  
*Ph.D in Electrical Engineering* Mar. 2020 – Feb. 2024

**Hanyang University** Seoul, Korea  
*Ph.D student, Department of Computer Software* Sep. 2018 – Feb. 2020

**Hanyang University** Seoul, Korea  
*MS in Computer Science* Mar. 2016 – Aug. 2018

**The Korean Academic Credit Bank System** Seoul, Korea  
*BS in Information Security Engineering* Mar. 2013 – Aug. 2015

## EXPERIENCE

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**Postdoctoral Researcher** Sep. 2024 – Present  
*University of Wisconsin-Madison* WI, USA

**Postdoctoral Researcher** Mar. 2024 – Aug. 2024  
*Korea Advanced Institute of Science and Technologies (KAIST)* Daejeon, Korea

## PUBLICATIONS

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### Conferences

1. [USENIX NSDI '26] Chengdong Wang, Joontaek Oh, Ming Liu, “Switched or Switchless: An Empirical Study of SAN Architecture for Disaggregated Storage”, In Proc. of 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI) 2026, May. 4-6, 2026
2. [ACM SAC '25] Juwon Kim, Dongeon Kim, Seung Won Yoo, Myoengin Cheon, Joontaek Oh, and Youjip Won, “lwFSCK: Light-weight Filesystem Check”, ACM Symposium on Applied Computing (SAC) 2025, March 31 - April 4, 2025
3. [USENIX FAST '25] Jieun Kim, Joontaek Oh, Juwon Kim, Seung Won Yoo, and Youjip Won, “OPIMQ: Order Preserving IO stack for Multi-Queue Block Device”, In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2025, Feb. 25-27, 2025

4. [USENIX FAST '25] Juwon Kim, Seungjae Lee, [Joontaek Oh](#), Dongkun Shin, and Youjip Won, "D2FS: Device-Driven Filesystem Garbage Collection", In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2025, Feb. 25-27, 2025
5. [USENIX FAST '25] Seung Won Yoo, [Joontaek Oh](#), Myoengin Cheon, Bonmoo Koo, Wonseob Jeong, Hyunsub Song, Hyeonho Song, Donghun Lee, and Youjip Won, "DJFS : Directory-Granularity Filesystem Journaling for Memory-Semantic SSD", In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2025, Feb. 25-27, 2025
6. [IEEE NVMSA '24] Seungho Lim, Seung Won Yoo, [Joontaek Oh](#), Wonseob Jeong, Hyunsub Song, Hyeonho Song, Donghun Lee, and Youjip Won, "Key-Space Partitioned LSM Tree for CMM-H", In Proc. of IEEE Non-Volatile Memory Systems and Applications Symposium (NVMSA) 2024, Aug 21 – 23, 2024
7. [USENIX FAST '23] [Joontaek Oh](#), Seung Won Yoo, Hojin Nam, Changwoo Min and Youjip Won, "CJFS : Concurrent Journaling for Better Scalability", In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2023, Feb. 20-23, 2022
8. [USENIX ATC '22] Juwon Kim, Minsu Jang, Danish Muhammad Teeshen, [Joontaek Oh](#), and Youjip Won "IPLFS: Log-Structured File System without Garbage Collection", In Proc. of USENIX Annual Technical Conference (ATC) 2022, July. 11-13, 2022
9. [ACM SYSTOR '22] Seung Won Yoo, [Joontaek Oh](#), and Youjip Won "O-AFA: Order Preserving All Flash Array", in Proc. of The ACM International Systems and Storage Conference (SYSTOR), Haifa, Israel, June. 13-15, 2022
10. [USENIX FAST '22] Dohyun Kim, Kwangwon Min, [Joontaek Oh](#), and Youjip Won "ScaleXFS: Getting scalability of XFS back on the ring", In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2022, Feb. 22-24, 2022
11. [USENIX FAST '22] [Joontaek Oh](#), Sion Ji, Yongjin Kim, and Youjip Won, "exF2FS: Transaction Support in Log-Structured Filesystem", In Proc. of USENIX Conference on File and Storage Technologies (FAST) 2022, Feb. 22-24, 2022
12. [ICISS '19] Myeongseon Kim, [Joontaek Oh](#), Youjip won, "Barrier enabled QEMU", In Proc. of ICISS 2019, Tokyo, Japan, Mar. 2019
13. [USENIX FAST '18] Youjip Won, Jaemin Jung, Gyeongyeol Choi, [Joontaek Oh](#), Seongbae Son, Jooyoung Hwang, Sangyeun Cho "Barrier Enabled IO Stack for Flash Storage", in proc. of USENIX Conference on File and Storage Technologies (FAST), Oakland, CA, USA, Feb. 12-15, 2018 (Awarded Best Paper)

## Journals

1. [ACM TOS] Youjip Won, [Joontaek Oh](#), Jaemin Jung, Gyeongyeol Choi, Seongbae Son, Jooyoung Hwang, Sangyeun Cho "Bringing Order to Chaos: Barrier-Enabled I/O Stack for Flash Storage", ACM Transactions on Storage (TOS)
2. [ACM TOS] Jinsoo Yoo, [Joontaek Oh](#), Seongjin Lee, Youjip Won, Jin-Yong Ha, Jongsung Lee, Junseok Shim, "OrcFS: Orchestrated File System for Flash Storage", ACM Transactions on Storage (TOS), Vol. 14, Issue 2, Apr, 2018

## Posters and Workshops

1. [ACM HotNet '25] Seunghyun An, [Joontaek Oh](#), Ming Liu, "Server Chiplet Networking", In Proc. of 24th ACM Workshop on Hot Topics in Networks (HotNets), 2025
2. [ACM APSys '21] Kyoungcho Koo, [Joontaek Oh](#), Kwangwon Min, Youngjin Kwon, Youjip Won, "C2J: Compulsory Compound Transaction for Journaling Filesystem", In Proc. of ACM SIGOPS Asia-Pacific Workshop on Systems (APSys), 2021
3. [USENIX FAST '19] [Joontaek Oh](#), Wonjong Lee, Youjip Won, "xF2FS: Supporting Multi-File Transaction in Log-Structured Filesystem", In Proc. of 17th USENIX Conference on File and Storage Technologies (FAST), 2019

4. [IEEE NVMSA '18] Joontaek Oh, and Youjip Won. “Embedded DBMS Design for In-Vehicle Information Management.” 2018 IEEE 7th Non-Volatile Memory Systems and Applications Symposium (NVMSA). IEEE, 2018.
5. [USENIX FAST '16] Jinsoo Yoo, Joontaek Oh, and Youjip Won. “Preserving Bi-Modal Utilization for Segment Cleaning in Modern Log-Structured Filesystem”, In Proc. of 14th USENIX Conference on File and Storage Technologies (FAST), 2016

## Patents

1. Youjip Won and Joontaek Oh, ”Commit block structure and device, for multiple file transaction”, US12346223B2, July 2025
2. Youjip Won and Joontaek Oh, ”On-disk data structure for commit and method of commit with the data structure in log-structured filesystem”, KR20220074807A, June 2022
3. Youjip Won and Joontaek Oh, ”Method to solve the problem that file operation is blocked because of journal conflict”, KR20220074804A, June 2022
4. Youjip Won and Joontaek Oh, ”Method and in-memory structure for a file operation processing multiple files”, KR20220074806A, June 2022
5. Youjip Won and Joontaek Oh, ”Method and apparatus for sending barrier command using dummy IO request”, KR20190096838A, Aug. 2019
6. Youjip Won and Joontaek Oh, ”Method and apparatus for parallel journaling using conflict page list”, KR20190096837A, Aug. 2019

## PROJECTS

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|                                                                          |                       |
|--------------------------------------------------------------------------|-----------------------|
| <b>High-Performance Exabyte Storage Systems</b>   Samsung Electronics    | May 2022 – May 2024   |
| <b>SNU-SKH Solution Research Center</b>   SK Hynix                       | Sep. 2021 – Aug. 2026 |
| <b>Disaggregated Memory System for Data Center</b>   Samsung Electronics | July 2021 – June 2021 |
| <b>Virtualization on Embedded Architecture</b>   Samsung Electronics     | Jan. 2021 – Dec. 2021 |
| <b>Lock-free Scalable IO Subsystem Design</b>   NRF                      | Jan. 2020 – Dec. 2025 |
| <b>Optimization of Filesystem and DBMS for Data Center</b>   Naver       | June 2019 – May 2020  |
| <b>Barrier-Enabled IO Stack for NVMe SSD</b>   SK Hynix                  | Nov. 2018 – Oct. 2019 |
| <b>Future Scalable OS</b>   IITP                                         | June 2018 – May 2023  |
| <b>Scalable IO Stack for future storage system</b>   NRF                 | June 2017 – Mar. 2020 |
| <b>System Software for Byte Addressable NVM</b>   KEIT/MOTIE             | Mar. 2016 – May 2017  |

## AWARDS AND HONORS

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|---------------------------------------------------|-----------|
| <b>Best Ph.D. student Award</b>   EE Dept., KAIST | Apr. 2023 |
| <b>Best TA Award</b>   EE Dept., KAIST            | Oct. 2020 |
| <b>Best Paper Award</b>   USENIX FAST 2018        | Feb. 2018 |

## TECHNICAL SKILLS

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**Languages:** C/C++, Python, SQL (MySQL, SQLite), R

**Developer Tools:** gcc/g++, gdb, Git, QEMU, gnuplot, vim, Emacs

**System knowledge:** In-depth knowledge in Linux kernel and EXT4, F2FS, XFS, Ceph, Linux-MD, etc.

## INVITED TALKS

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**File Systems and I/O Stacks for Emerging Storage Devices**

*Gyeongsang National University, Jun. 12. 2025*

**Separating the Ordering Guarantee and Durability Guarantee in Modern IO Stack**

*Georgia Institute of Technology, Apr. 24. 2024*

**IO Stack Design for Better Scalability**

*IBM Research, May. 23. 2023*

## SERVICE

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Journal Reviewer, ACM Transactions on Storage (TOS).

Sep. 2025

## TEACHING

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**TA, Programming Structure for EE (EE 209)**

*Korea Advanced Institute of Science and Technologies (KAIST)*

Mar. 2021 – June 2021

*Daejeon, Korea*

**TA, Introduction to Operating Systems (EE 415)**

*Korea Advanced Institute of Science and Technologies (KAIST)*

Sep. 2020 – Dec. 2020

*Daejeon, Korea*

**TA, Commissioned Education of IO Subsystem**

*Samsung Advanced Technology Training Institute*

July 2020

*Suwon-si, Korea*

**TA, Unix Kernel Design (EE 488)**

*Korea Advanced Institute of Science and Technologies (KAIST)*

Mar. 2020 – June 2020

*Daejeon, Korea*

**TA, SK Hynix - KAIST ASK Program**

*Korea Advanced Institute of Science and Technologies (KAIST)*

Feb. 2020

*Daejeon, Korea*

**TA, Introduction to Operating Systems (EE 415)**

*Korea Advanced Institute of Science and Technologies (KAIST)*

Sep. 2019 – Dec. 2019

*Daejeon, Korea*

**TA, Commissioned Education of IO Subsystem**

*Samsung Advanced Technology Training Institute*

June 2019

*Suwon-si, Korea*

**TA, Unix Kernel Design (EE 488)**

*Korea Advanced Institute of Science and Technologies (KAIST)*

Mar. 2019 – June 2019

*Daejeon, Korea*

**TA, System Programming (CSE 4009)**

*Hanyang University*

Sep. 2018 – Dec. 2018

*Seoul, Korea*

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|------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>TA, Commissioned Education of IO Subsystem</b><br><i>Samsung Advanced Technology Training Institute</i> | Aug. 2018<br><i>Suwon-si, Korea</i>          |
| <b>TA, Operating System (ELE 3021)</b><br><i>Hanyang University</i>                                        | Mar. 2018 – June 2018<br><i>Seoul, Korea</i> |
| <b>TA, System Programming (CSE 4009)</b><br><i>Hanyang University</i>                                      | Sep. 2017 – Dec. 2017<br><i>Seoul, Korea</i> |
| <b>TA, Operating Systems &amp; System Programming (ITE 2032)</b><br><i>Hanyang University</i>              | Sep. 2016 – Dec. 2016<br><i>Seoul, Korea</i> |
| <b>TA, Commissioned Education of Operating System</b><br><i>Samsung Electronics</i>                        | Aug. 2016<br><i>Suwon-si, Korea</i>          |
| <b>TA, Introduction to Operating System (ELE 3021)</b><br><i>Hanyang University</i>                        | Mar. 2016 – June 2016<br><i>Seoul, Korea</i> |

## REFERENCES

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|                                                          |                                        |
|----------------------------------------------------------|----------------------------------------|
| <b>Prof. Youjip Won</b>   ywon@kaist.ac.kr               | Dept. of Electrical Engineering, KAIST |
| <b>Prof. Youngjin Kwon</b>   yjkwon@kaist.ac.kr          | School of Computing, KAIST             |
| <b>Dr. Changwoo Min</b>   changwoo@igalia.com            | Igalia                                 |
| <b>Prof. Ming Liu</b>   mgliu@cs.wisc.edu                | Dept. of CS, UW-Madison                |
| <b>Prof. Andrea Arpaci-Dusseau</b>   dusseau@cs.wisc.edu | Dept. of CS, UW-Madison                |
| <b>Prof. Remzi Arpaci-Dusseau</b>   remzi@cs.wisc.edu    | Dept. of CS, UW-Madison                |